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NETFLIX EPISODE COMPLETION ANALYSIS AND MODELING REPORT

INTRODUCTION

In this analysis, we delve into a comprehensive dataset from Netflix, a leading streaming service, to explore user interactions and viewing patterns. The dataset includes vital details such as user IDs, viewing dates and times, show titles, seasons, episodes, time watched, user gender, and episode completion status. Our primary aim is to clean and explore this data, addressing any missing values and forming hypotheses about factors influencing episode completion. We then build and compare various predictive models — including Multiple Logistic Regression with all variables, a PCA-reduced Logistic Regression, and a Decision Tree model — to identify key determinants of episode completion. This analysis is geared towards providing actionable insights and strategic recommendations to Netflix. By understanding what drives users to complete episodes, Netflix can enhance user engagement and optimize content recommendations, ultimately refining their content strategy and improving viewer satisfaction.

DATA CLEANSING

While approaching the Netflix data we found potential missing data in numerous columns as follows:

35 records were missing in Season column
18 records were missing in Episode column
17 records were missing in Time Watched column
99 records were missing in Gender column
47 records were missing in Completed column
100 records were missing in Time-of-day column

We handled missing data by replacing the numeric variables with median and categorical variables with mode. We have taken this approach due to the following reasons:

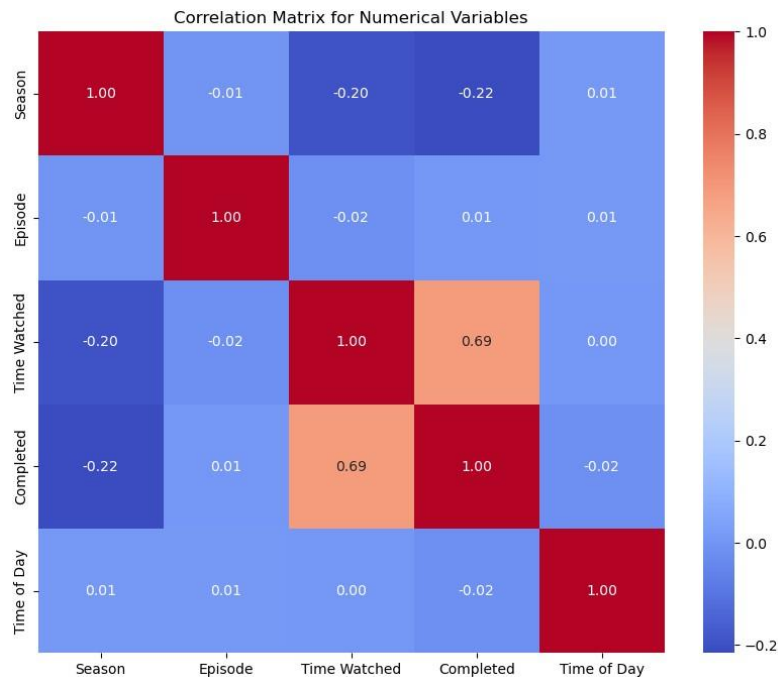
- **Robustness to Outliers:** The median is less sensitive to outliers compared to the mean. If the data distribution is skewed or contains outliers, using the mean can lead to biased imputation. The median, being the middle value, provides a more robust estimate for the central tendency of the data.
- **Maintaining Data Integrity:** Imputing missing values allows us to retain data rows that would otherwise be discarded if we chose to delete rows with missing values. This is particularly important when the dataset is not very large, as discarding data could lead to loss of valuable information.
- **Suitability for Ordinal and Discrete Data:** For data that is ordinal (having a natural order) or discrete (like 'Season' or 'Episode' in the Netflix dataset), the median can provide a more meaningful imputation compared to the mean, which might result in a non-existent or non-intuitive value.

- **Statistical Stability:** The median, being a measure of central tendency, provides a statistically stable and representative value for imputation, especially when the distribution of the data is not known or when the data is not normally distributed.

In the context of the Netflix dataset, using the median helps in dealing with potential skewness in the distribution of variables like 'Season', 'Episode', and 'Time Watched', ensuring that the imputed values do not distort the dataset's original structure significantly.

DATA EXPLORATION

CORRELATION PLOT



The provided correlation matrix for numerical variables in the dataset shows the relationships between different factors such as 'Season', 'Episode', 'Time Watched', 'Completed', and 'Time of Day'. A strong positive correlation (0.69) is observed between 'Time Watched' and 'Completed', suggesting that the longer a user watches an episode, the more likely they are to complete it. Negative correlations with 'Completed' for 'Season' (-0.22) and 'Time of Day' (-0.02) suggest that later seasons and certain times of day might have a slight negative impact on the likelihood of episode completion. However, these negative correlations are weak, indicating that these factors have a minimal linear relationship with episode completion. The near-zero correlations between 'Episode' and other variables imply that the episode number has very little linear association with both time watched and completion rates. Overall, the matrix provides insights into which factors might be worth investigating further for their impact on a viewer's likelihood to complete an episode.

HYPOTHESES

To understand episode completion factors, the following hypotheses can be explored:

- **Time Watched:** There is a direct relationship between the amount of time watched and the likelihood of episode completion.
- **Time of Day:** The time of day when an episode is watched might influence completion rates.
- **Season and Episode:** The position of an episode within a season (early vs. late) could impact completion likelihood.

Hypothesis 1: Time Watched

- **Alternative Hypothesis (H1):** There is a significant relationship between the amount of time a user watches an episode and the likelihood of episode completion. Specifically, longer viewing times are associated with a higher probability of episode completion.
- **Null Hypothesis (H0):** The amount of time a user watches an episode has no significant relationship with the likelihood of episode completion.

Hypothesis 2: Time of Day

- **Alternative Hypothesis (H1):** The time of day when an episode is watched significantly influences the likelihood of completion. Certain times of day might see higher completion rates due to factors like viewer availability or preferences.
- **Null Hypothesis (H0):** The time of day when an episode is watched does not significantly influence the likelihood of completion.

Hypothesis 3: Episode Characteristics

- **Alternative Hypothesis (H1):** Specific attributes of episodes, such as their position within a season (early vs. late) or the season number, significantly affect the likelihood of episode completion. This hypothesis is based on the idea that viewer engagement might vary across different stages of a show's storyline.
- **Null Hypothesis (H0):** The position of an episode within a season or the season number does not significantly affect the likelihood of episode completion.

The alternative hypotheses (H1) represent our assumptions or predictions about the dataset, suggesting that there are significant relationships between the proposed variables and episode completion rates. The null hypotheses (H0), on the other hand, posit that there are no such relationships. Through our analysis and model building, we aim to test these hypotheses, seeking evidence to either reject the null hypotheses and support the alternative hypotheses, or fail to reject the null, suggesting that more investigation is needed.

MODELLING

Logistic Regression

logistic regression was performed with all the variables including categorical variables by applying one hot encoding

Logistic Regression with All Variables

Precision: The model is precise in its predictions, with a high score (0.95 for class 0 and 0.92 for class 1), indicating that the majority of its predictions for both classes are correct.

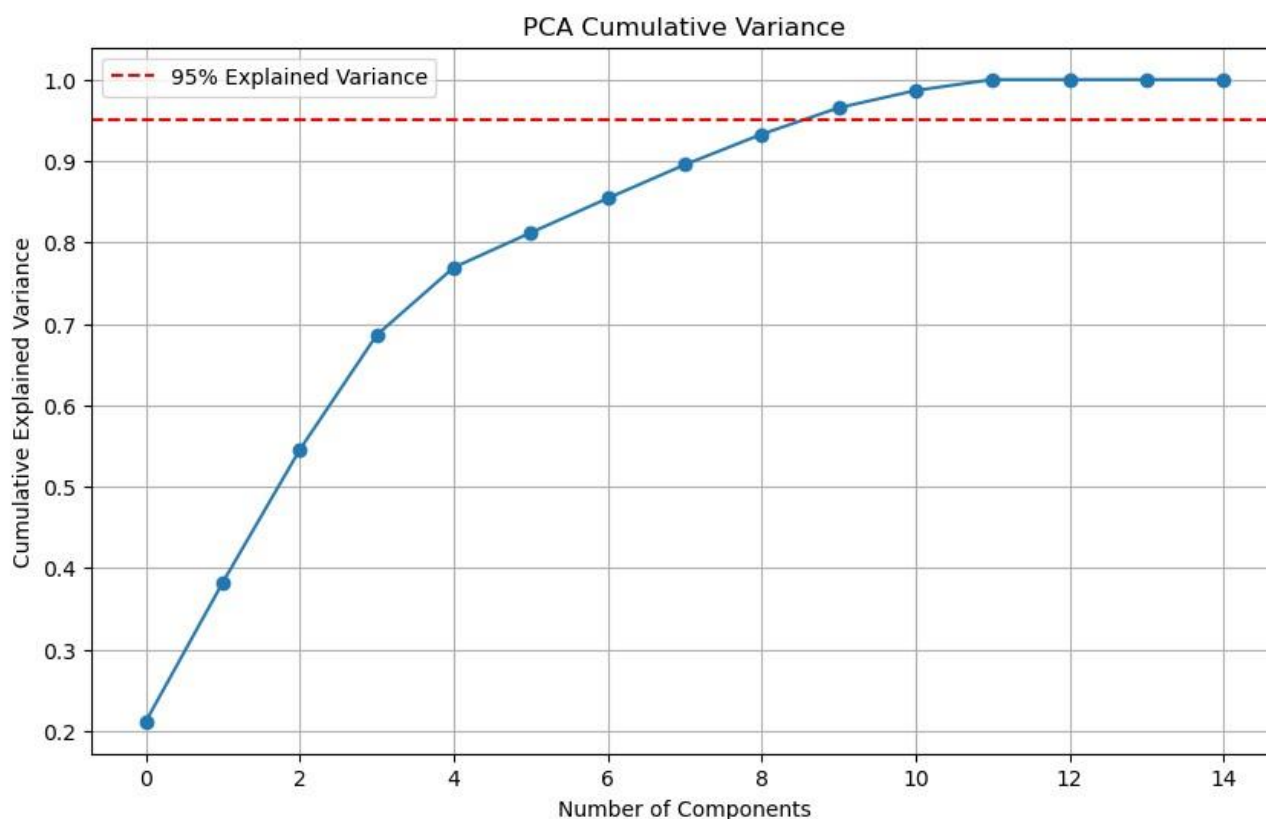
Recall: This model is also highly sensitive, correctly identifying 97% of all true class 0 instances and 86% of all true class 1 instances.

F1-Score: The F1-scores are high for both classes (0.96 for class 0 and 0.89 for class 1), indicating a good balance between precision and recall.

Accuracy: With an overall accuracy of 94%, this model is very effective at predicting episode completion.

Logistic Regression with PCA

Before performing logistic regression we have found the cumulative variance for the components to understand the optimal number of PCA components below is the plot:



This PCA analysis suggests that the dataset can be effectively reduced to about 9 or 10 dimensions (principal components) without losing significant information. This dimensionality reduction can simplify the modeling process and potentially improve the efficiency and generalizability of the predictive models. Below are the results after performing logistic regression with PCA variables.

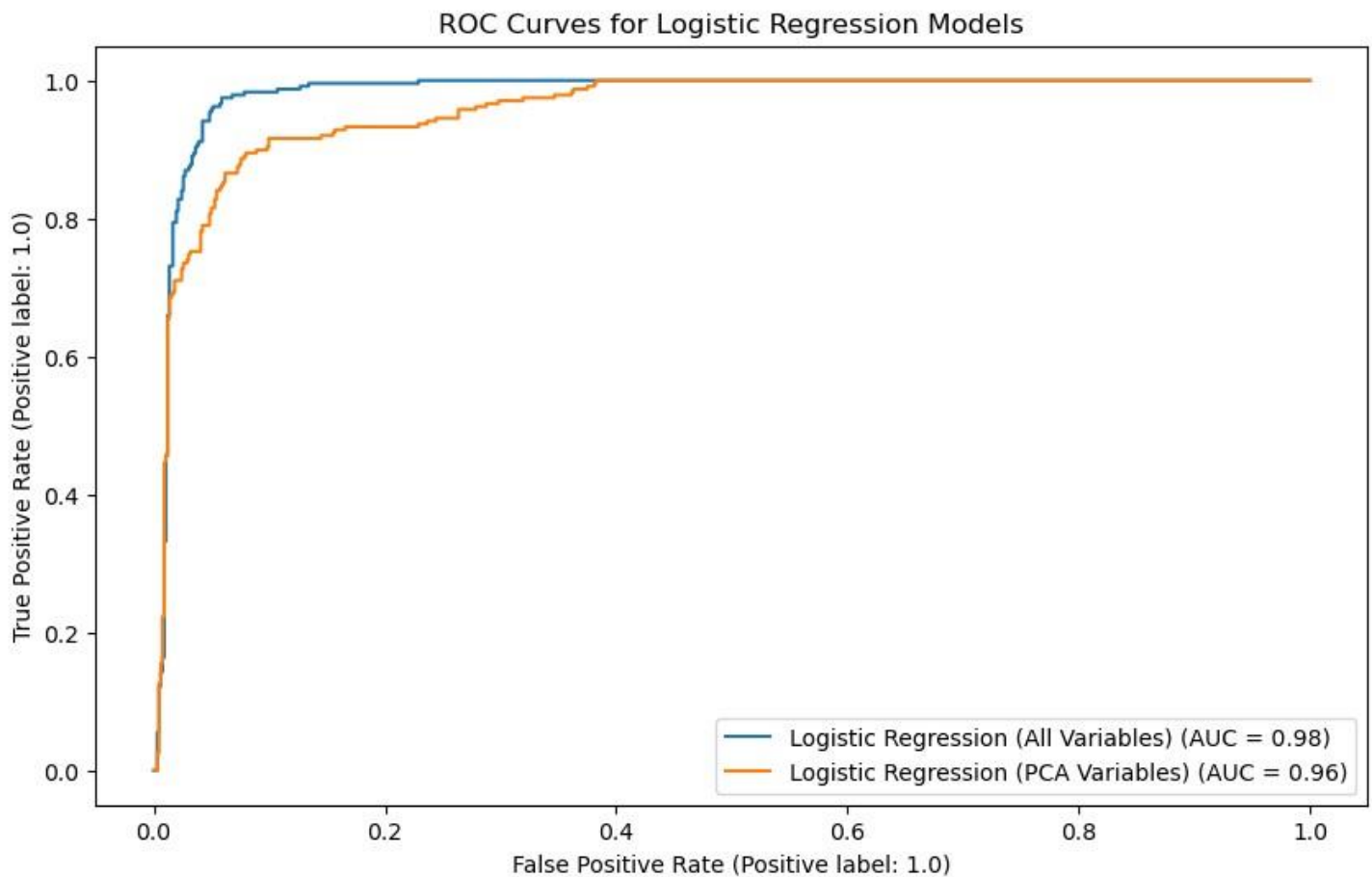
Precision: There is a slight decrease in precision for both classes compared to the first model, especially for class 1 (0.87), which indicates that the PCA model makes more false positive predictions for episode completion.

Recall: The recall for class 1 (0.79) shows a more considerable drop, indicating that this model is less sensitive in identifying true completions compared to the first model.

F1-Score: The F1-scores are lower than the first model, reflecting the decrease in both precision and recall, especially for class 1.

Accuracy: The overall accuracy of 91% is lower than the first model, which suggests that dimensionality reduction via PCA might have removed some information that was useful for prediction

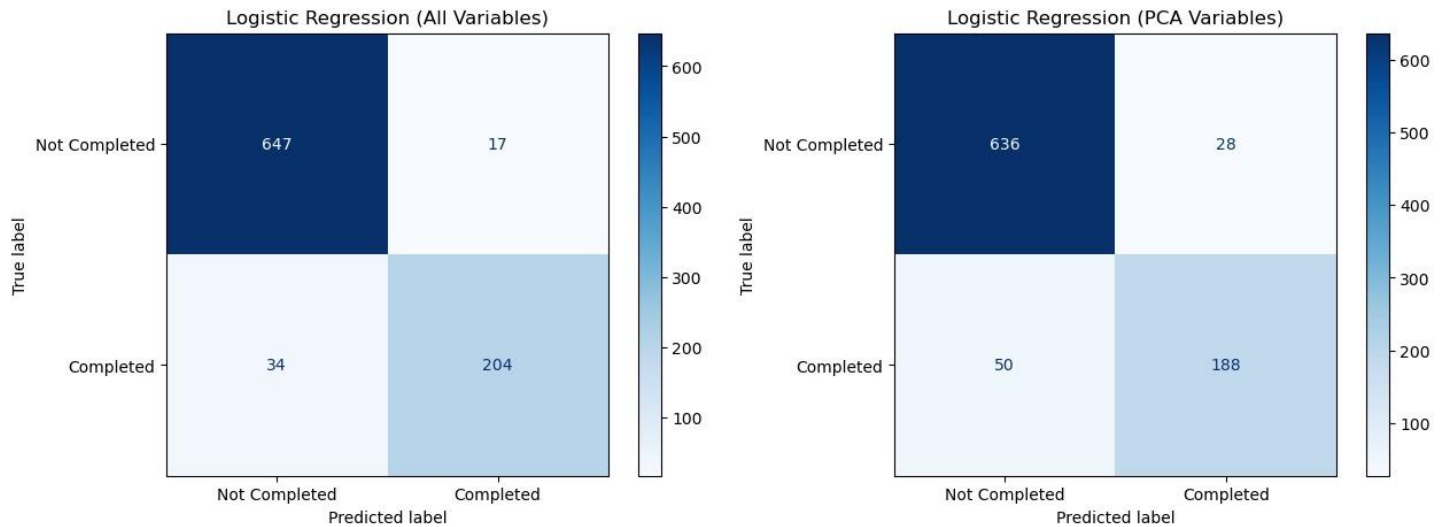
ROC CURVES FOR LOGISTIC REGRESSION MODELS



The image depicts ROC curves for two logistic regression models, one using all variables (blue line) and the other using variables reduced by PCA (orange line). The ROC curve is a plot of the true positive rate (sensitivity) against the false positive rate (1-specificity) for the different possible cut-points of a diagnostic test. The area under the curve (AUC) provides a measure of the model's ability to distinguish between the positive and negative classes. A higher AUC indicates better model performance. In this case, the Logistic Regression model using all variables has an AUC of 0.98, indicating a very high ability to discriminate between episodes that were completed and not completed. The PCA-based model has a

slightly lower AUC of 0.96, which still indicates high discriminative power, but slightly less than the model using all variables. Both models demonstrate excellent performance, with the full model having a slight edge, possibly due to retaining more informative features that were lost during PCA's dimensionality reduction.

Confusion Matrix for Regression models

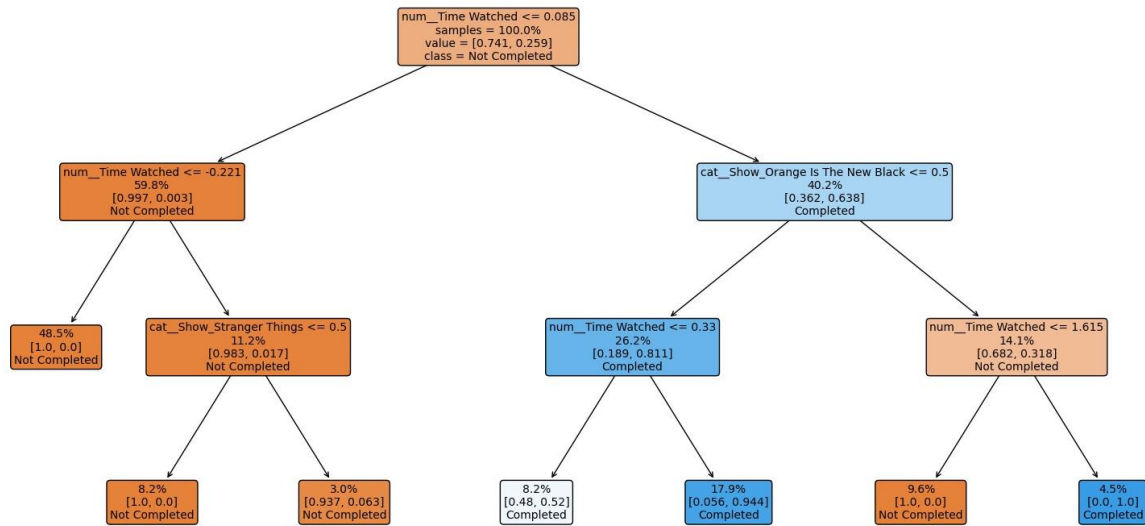


The image shows confusion matrices for two logistic regression models. The first matrix, corresponding to the model using all variables, indicates a high number of true negatives (TN) and true positives (TP), with relatively few false negatives (FN) and false positives (FP), which suggests that the model is performing well at classifying both completed and not completed episodes. The second matrix, for the logistic regression model with PCA variables, shows a slightly higher number of FNs and FPs, indicating a modest decrease in performance. This decrease may be due to the loss of information during the PCA process, which can affect the model's ability to accurately classify instances. Both models show strong performance, but the non-PCA model appears to have a better balance in correctly predicting both classes.

DECISION TREE

- **Precision:** The Decision Tree model has a very high precision, comparable to the first model, indicating that most of its predictions are correct for both classes.
- **Recall:** The recall is exceptionally high (0.99 for class 0 and 0.92 for class 1), indicating that this model is very good at identifying true instances of both classes.
- **F1-Score:** Reflecting the high precision and recall, the F1-scores are also very high, particularly for class 0 (0.98).
- **Accuracy:** With an overall accuracy of 97%, the Decision Tree model outperforms both logistic regression models.

Decision Tree Visualization



The decision tree visualization, representing the model's decision-making process for classifying episodes as completed or not completed. The tree structure is based on splitting the data using various features, with the splits representing the decision nodes and the end points representing the leaves, where final decisions are made. Starting from the top, the model first evaluates 'Time Watched', splitting the data into two branches. As we move down the tree, further splits are made based on additional features like 'Show'. The leaves provide the classification outcome, with the percentages indicating the proportion of samples in that leaf and the class prediction (Completed or Not Completed). The decision tree shows the hierarchical importance of features according to how the model makes predictions, revealing, for instance, that 'Time Watched' and certain 'Shows' are significant determinants in predicting episode completion. The colors in the nodes likely represent the purity of the class at that point, with one color representing 'Not Completed' and another 'Completed', the intensity of the color possibly indicating the confidence or proportion of the samples in that node.

Overview and Comparison

- The Logistic Regression model using all variables demonstrates strong predictive performance with high precision and recall across both classes.
- The Logistic Regression model with PCA, while still performing well, shows a decline in all metrics. This suggests that the PCA transformation, by reducing the feature space, may have discarded some relevant information that contributes to predicting episode completion.
- The Decision Tree model exhibits the best performance among the three models in terms of all evaluated metrics, including the highest accuracy. This might be due to its ability to capture non-linear relationships and interactions between features that logistic regression models cannot.

In summary, each model has its strengths, with the non-PCA Logistic Regression model offering a good balance of precision and recall, while the Decision Tree model shows the highest overall accuracy and performance. The PCA-based Logistic Regression model might benefit from tuning the number of components used in PCA to potentially improve its predictive power. These insights would be crucial for Netflix in developing strategies to increase viewer engagement by predicting and enhancing episode completion rates.

CONCLUSION

The analysis and modeling performed provide valuable insights into factors affecting episode completion on Netflix. The Decision Tree model, in particular, demonstrated a high predictive capability, suggesting a strong correlation between certain features and episode completion. These findings can guide strategic decisions for content recommendation and scheduling to enhance viewer engagement and satisfaction.